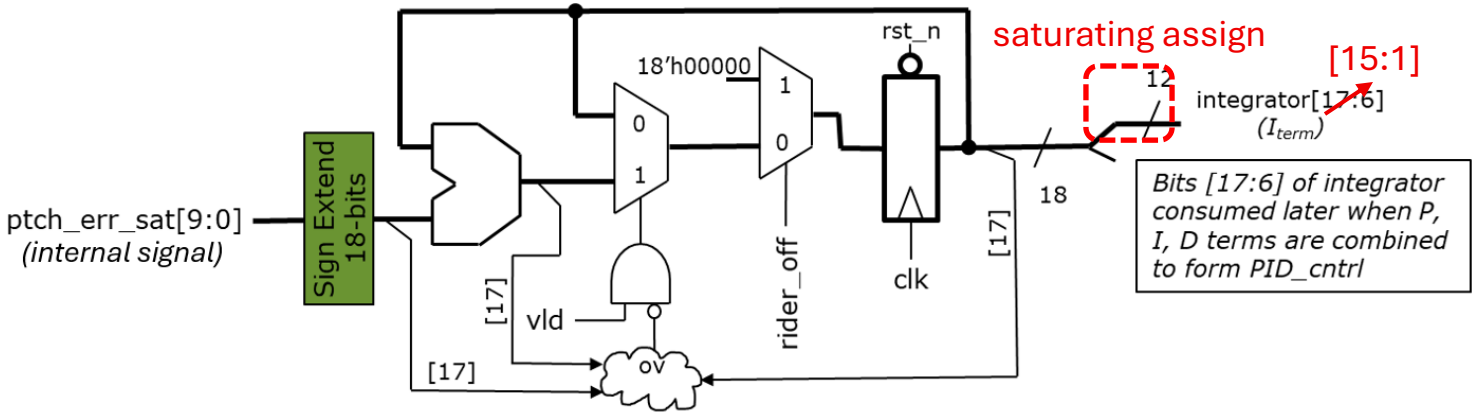
Exercise 14: fast_sim & balance_cntrl

- There are some long time constants in PID
  - **ss_tmr** ramps up over 2-seconds (100+ million clks)
  - **integrator** integrates slowly
- These values are good for actual hardware (FPGA) but take too long to simulate.

- We need a way of speeding up simulation. We will introduce a parameter called **fast_sim**
  - Default to 1 for simulation
  - Set to 0 when we actually map to FPGA in the project demo
  - You will use **generate conditional** to modify how things are done in simulation vs hardware
    - We want to introduce minimal change in terms of structure / design so our simulation is still useful

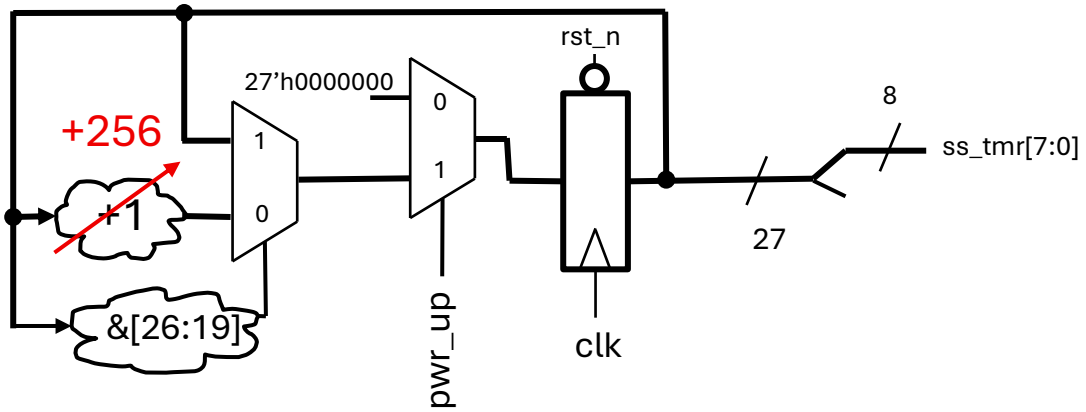
Introducing fast_sim to PID

- To speed up **integrator**
  - As is, **I_term** is tapped from bits [17:6] of the integrator accumulator, then sign extended when combining P, I, D
    → make **I_term** 15-bit instead: *{{3{integrator[17]}},integrator[17:6]};*
  - Under the condition of **fast_sim=1** we will now tap bits [15:1] to form **I_term**.
  - Not using bits [17:16] means there is danger that the integrator is too positive or too negative to represent in **I_term**, so this assignment under **fast_sim** should be a **saturating assign** that inspects bits [17:15] to determine if +/- saturation occurs



Introducing fast_sim to PID

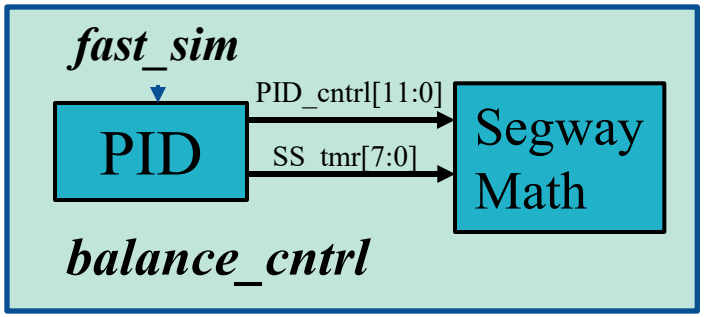
- To speed up **ss_tmr**
  - As is, we increment the 27-bit timer by 1 every clk
  - Under the condition of **fast_sim=1** we will now increment the 27-bit timer by 256 every clk



- There is a provided testbench: **PID_fastsim_tb.sv**
  - You can use it to check against your modified PID

Create balance_cntrl

- **balance_cntrl** is just a combination of **PID** and **SegwayMath**



Signal:	Dir	Description
clk,rst_n	in	50MHz system clock & active low reset
vld	in	High whenever new inertial sensor reading (**ptch**) is ready
ptch[15:0]	in	Pitch of Segway from **inertial_intf**
ptch_rt[15:0]	in	Pitch rate (degrees/sec). Used for D_term of PID
pwr_up	in	Asserted when Segway balance control is powered up. Used to keep ss_tmr at zero until then.
rider_off	in	Asserted when no rider detected. Zeros out integrator.
steer_pot[11:0]	in	From A2D_intf (converted from steering potentiometer)
en_steer	in	Enables steering control
lft_spd[11:0]	out	12-bit signed speed of left motor
rght_spd[11:0]	out	12-bit signed speed of left motor
too_fast	out	Rider approaching point of minimal control margin

Create balance_cntrl

- Create **balance_cntrl.sv**
  - Note the connections between **PID** and **SegwayMath** (**PID_cntrl[11:0]** and **ss_tmr[7:0]**)

- **fast_sim** should also be a parameter at the **balance_cntrl** level of hierarchy and should be defaulted to 1
- Compile **balance_cntrl.sv** in ModelSim so you know at least it is free of syntax errors.
  - We will test it in Exercise 16 so don't worry about it too much yet.

- Submit **balance_cntrl.sv**